

Appunti Completi del Corso

Il Tuo Nome

22 novembre 2025

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Capitolo 1

Week 13-19 October

1.0.1 Introduction

In most of the course we'll handle wireless communications, as they're much more difficult than wired communications. In this course we'll stay at the physical layer. We'll see:

- **Channel models:** taking radio communications and making a mathematical model for them. We'll find an accurate model and then we'll apply a tradeoff to it, simplifying it so that it is manageable.
- **OFDM:** Orthogonal Frequency Division Multiplexing.
- **MIMO:** we have multiple antennas at the transmitter and multiple antennas at the receiver; we'd like to see how much this improves communications. The book he selected this year handles fundamentals of MIMO communications. MIMO is the latest and greatest achievement, he chose a book which goes from the basics. We might have single user MIMO or even multiple user MIMO: we're communicating to multiple phones through multiple antennas.

We can ask ourselves: what is the best that we can do? This is what our approach will be, so that we can know the boundaries. We'll follow the slides from the author of the book, hopefully they'll be good. We'll also have exercises with solutions: we can look at them from cambridge.org, specifically from the page of the book.

1.0.2 Primer

The first chapter is a primer in information theory and MSE estimation; we'll go fast as it should summarize stuff we know, as mutual information between two variables. !!: the signals are always complex-valued and the constellations are always normalized to have unit power. ∞ -QAM: we increase the number of points making them denser inside a specific set.

Complex symmetric random variables, ∞ -modulations

We'll start from chapter 1 of the book (000): *primer of information theory and MSE*.

We should already know this stuff, but we'll cover it in order to have a common starting ground.

Prof's backstory moment: In the end, he studies **physical layer** and that's it.

We ask ourselves: “**what is information?** When can we say that someone is giving us information?”. *Professor begins ranting about information*: information is related to **probability!** The more unlikely something is, the more informative it is!

Today we'll talk about **mutual information** between two random variables: the information a variable can supply to another variable.

We'll talk about **channel capacity**: the capacity at which a channel can convey information; it answers to “how much information can I send through this channel, reliably?”. By reliably we mean that we can lower the probability of error as much as we want.

Binary coding and **decoding** are near optimum, but there can be other ways, which could be better: we'll show that binary is indeed **optimum**.

LMMSE is optimum when the signal and noise are **Gaussian**.

Let's start from **signals**; in the book, signals are normally **complex**, unless we specifically say otherwise. Normally **complex random variables** are **complexly symmetric**:

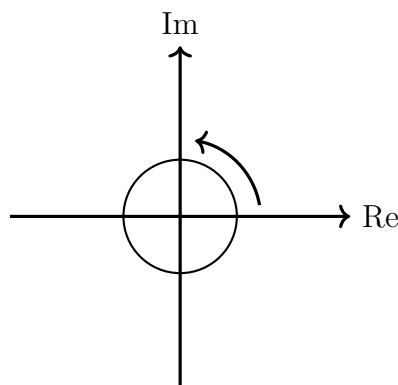
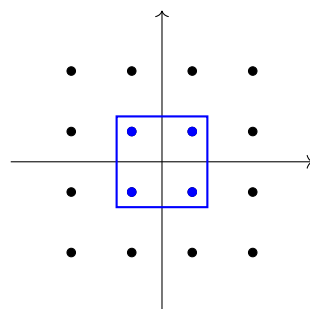


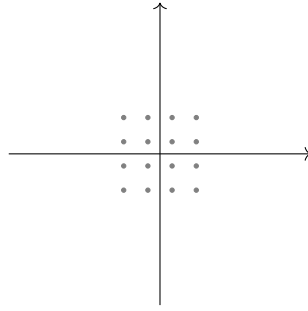
Figura 1.1: Rappresentazione del piano Im-Re con una PFD simmetrica centrale e freccia di rotazione.

About **M-PSK** and **M-QAM**: M is usually a **power of 2**.

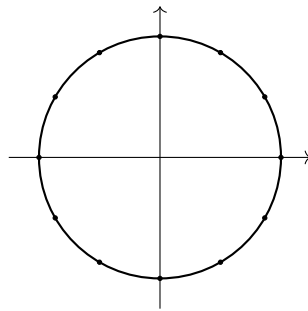
Let's see **16-QAM**:



The power of the ones in blue is lower than the power of the other signals. In the book they always keep the **same variance of the constellation**: the book squeezes the values so that the **power stays constant**.



The book also has **inf-PSK** and **inf-QAM**: We take the **normal constellation** and we keep **adding points**.



∞ – **PSK**

It's a useful way to see what happens in the **asymptotic regime**. Also, the **PDF** of a **complex Gaussian variable exists** (see notes).

If σ^2 is the variance, the standard deviation is $\sqrt{\sigma^2} = \sigma$ $\downarrow$ NOT $\pm \sigma$!	}
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We'll try and be as precise as possible: there is no **negative standard deviation**! As an example:

$$f(x) = \sqrt{x} \in \mathbb{R} \text{ is def. only for } x \geq 0$$

$$x^2 = 16 \text{ has two sol.}$$

What about information? There have been many results, dating back to **Shannon**! (See pag 7 of the slides btw)

Discrete entropy (basic, joint, conditional), proper signals (+example with the entropy of a Bernoulli random variable)

1.1 Entropy

1.1.1 Entropy of a discrete-time random variable

The **entropy** $\mathcal{H}(x)$ of a discrete-time random variable x is a nonnegative function that quantifies the amount of uncertainty in x .

It is defined as:

$$\mathcal{H}(x) = - \sum_x P(x) \log_2 P(x) = -\mathbb{E}[\log_2 P(x)] \quad (1.1)$$

1.1.2 Joint Entropy

The **Joint entropy** $\mathcal{H}(x_0, x_1)$ is used in statistical physics and is defined as:

$$\mathcal{H}(x_0, x_1) = - \sum_{x_0} \sum_{x_1} p_{x_0 x_1}(x_0, x_1) \log_2 p_{x_0 x_1}(x_0, x_1) = -\mathbb{E}[\log_2 p_{x_0 x_1}(x_0, x_1)] \quad (1.2)$$

(Note: Differential information can be negative)

1.1.3 Conditional Entropy

The **Conditional entropy** $\mathcal{H}(x|y)$ is defined as:

$$\mathcal{H}(x|y) = - \sum_x \sum_y p_{xy}(x, y) \log_2 p_{x|y}(x|y) \quad (1.3)$$

1.1.4 Entropy Connections

The joint entropy can be related to conditional and marginal entropies using the chain rule:

$$\mathcal{H}(x, y) = \mathcal{H}(x) + \mathcal{H}(y|x) = \mathcal{H}(y) + \mathcal{H}(x|y) \quad (1.4)$$

1.2 Example: Entropy of a Bernoulli Random Variable

Consider a **Bernoulli random variable** $X = \{0, 1\}$ with probabilities $P = \{p, 1 - p\}$.

The entropy $H(X)$ will be ≤ 1 . The equality holds if $p = 1/2$.

- If the coin always flips heads or tails (i.e., $p = 0$ or $p = 1$), we get **no information** ($H(X) = 0$).
- We get **maximum information** ($H(X) = 1$) when we flip a *fair* coin ($p = 1/2$).
- One could say that the messier the information, the most we get out of it.

The plot of $H(x)$ versus P is a concave function, maximized at $P = 1/2$.

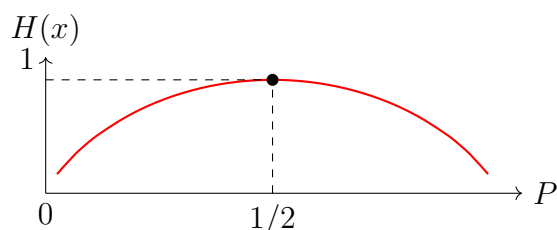


Figura 1.2: Grafico di $H(x)$ vs P per una variabile Bernoulli (massimo a $P=1/2$).

1.3 Proper Complex Signals

The course always deals with “**proper**” signals.

For proper signals, the **real and imaginary part** of those numbers are **uncorrelated**. Properness is preserved by affine transformations.

An intuitive reason to consider only proper complex numbers is to avoid cases like the following complex variable:

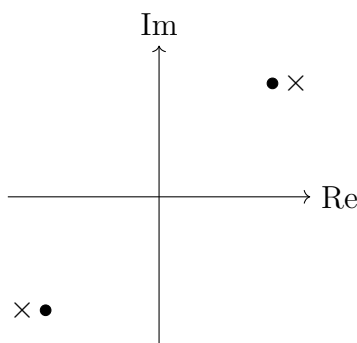


Figura 1.3: Esempio di una variabile complessa non 'proper'.

1.3.1 Definition of Properness

A complex random variable is said to be **proper** if the condition $\mathbb{E}[x^2] = \mathbb{E}[x]^2$ holds.

- Properness requires $\text{Re}[x]$ and $\text{Im}[x]$ to be **uncorrelated** and have the **same variance**.

A complex random vector is said to be proper if $\mathbb{E}[\mathbf{x}\mathbf{x}^T] = \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^T$ holds. This is equivalent to asking that $\text{Re}[\mathbf{x}]$ and $\text{Im}[\mathbf{x}]$ have identical covariance matrices and cross-covariance equal to $\mathbf{0}$.

1.4 Entropy Rate

If we have a sequence of discrete random variables indexed by time, $x_0, \dots, x_{N-1}$, and they are **IID** (Independent and Identically Distributed), the entropy of the whole process grows linearly with N at a rate of $\mathcal{H}(x_0)$.

In general, a process grows according to the **entropy rate**, $\mathcal{H}$, defined as:

$$\mathcal{H} = \lim_{N \rightarrow +\infty} \frac{1}{N} \mathcal{H}(x_0, \dots, x_{N-1}) \quad (1.5)$$

If the process is stationary, the entropy rate is also equal to:

$$\mathcal{H} = \lim_{N \rightarrow +\infty} \mathcal{H}(x_N | x_0, \dots, x_{N-1}) \quad (1.6)$$

This concept can be extended to continuous processes as **differential entropy** ($\mathfrak{h}$).

- A process is said to be **nonregular** if its present value is perfectly predictable from noiseless observations of the entire past.
- A process is **regular** (which happens if $\mathfrak{h} > -\infty$) if its present value cannot be perfectly predicted from noiseless observations of the entire past.

1.5 Differential Entropy

The next topic covers the **differential entropy** of a **Complex Gaussian scalar** and a **Complex Gaussian vector**.